



Byungjun Kim

M.S. Student
Graduate School of AI
Korea Advanced Institute of Science and Technology,
Daejeon, South Korea

+82 10-5001-9430

bjkim@kaist.ac.kr

www.linkedin.com/in/byungjunk

EDUCATION

•Korea Advanced Institute of Science and Technology, Daejeon

2025 ~ Present

M.S., Graduate School of AI
Bio-Imaging Signal Processing Lab (BISPL)

•Korea Advanced Institute of Science and Technology, Daejeon

2019 ~ 2025

B.S., School of Electrical Engineering

GPA: 4.12/4.3 (Summa cum laude)

EXPERIENCE

•Barreleye Inc.

Jan. 2024 ~ Aug. 2024

Internship

Seoul, South Korea

– Development of BI-RADS Compliant Breast Lesion Identification Algorithms

Engineered morphological features and optimized machine learning models, achieving high accuracy in shape classification by quantifying tumor irregularities.

– Lightweight Deep Learning Architecture Design & Medical Device Integration

Designed a resource-efficient backbone network using Squeeze-and-Excitation layers and Atrous Convolution Blocks, successfully deploying the prototype onto a medical device for real-time inference.

PUBLICATIONS

•Diverse Text-to-Image Generation via Contrastive Noise Optimization

ICLR (2026)

Byungjun Kim*, Soobin Um*, Jong Chul Ye†

- Spearheaded the development of **Contrastive Noise Optimization (CNO)**, a novel inference-time framework that mitigates mode collapse in diffusion models by optimizing initial noise vectors via an objective min/max interpretation.
- Designed a contrastive loss objective operating in the denoised space to maximize semantic diversity while preserving fidelity, validated through theoretical analysis of mutual information bounds.
- Achieved **State-of-the-Art (SOTA)** performance on the quality-diversity Pareto frontier across various backbones (SDXL, SD3, FLUX) with minimal computational overhead.

•MotionCFG: Boosting Motion Dynamics via Stochastic Concept Perturbation

Under Review (2026)

Byungjun Kim, Soobin Um†, Jong Chul Ye†

- Proposed **MotionCFG**, a training-free guidance framework that resolves the inherent static bias in Text-to-Video (T2V) diffusion models by contrasting target concepts with noise-perturbed semantic counterparts.
- Effectively mitigated **Content-Motion Drift** through implicit hard negative mining, injecting partial embedding perturbations to refine temporal details without compromising global spatial integrity.
- Demonstrated superior temporal dynamics across state-of-the-art models (Wan2.1, CogVideoX) and successfully extended the mechanism to steer precise object numerosity (counting).

AWARDS & SCHOLARSHIPS

•Academic Awards

Dean's List in Spring 2021, KAIST
Dean's List in Fall 2023, KAIST

•Scholarships

National Excellence Scholarship (Natural Sciences and Engineering)

FIELDS OF INTEREST

- **Deep Generative Models:** Diffusion Models, Flow Matching, Controllable Synthesis, Multimodal Generation
- **Video Generation:** Autoregressive Video Generation, Text/Image-to-Video Synthesis, Long-Horizon Temporal Modeling, Motion Dynamics
- **Inference-Time Guidance:** Training-Free Sampling, Classifier-Free Guidance (CFG), Noise/Embedding Optimization, Adaptive Guidance Scheduling